

Deep

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Education

Thapar Institute of Engineering & Technology, Patiala
Master of Computer Applications (MCA)

Aug 2024 – May 2026
CGPA: 8.53

Kurukshetra University, Kurukshetra
Bachelor of Science (B.Sc.)

Aug 2021 – July 2024
Percentage: 64.2%

Experience

Trainee Software Engineer (Intern) | Mphasis Limited

Feb 2026 – May 2026 — [Link](#)

- Engineered QuantumLogic Labs, a full-stack AI-powered Applicant Tracking System using FastAPI, React, PostgreSQL, featuring dual-parser resume extraction, contact-safe profile parsing, and hybrid candidate scoring using weighted skill taxonomy, semantic similarity, impact analysis, and experience relevance.
- Built an end-to-end hiring workflow with public candidate intake, Google OAuth2 authentication, JWT-based role management, automated email notifications, recruiter chatbot, JD optimizer, interview scheduling, analytics dashboard, and immutable audit logging across a normalized PostgreSQL schema.
- Developed a recruiter intelligence layer supporting hiring decisions (Strong Hire / Hold / Reject), pipeline stage tracking, skill-gap analysis, resume quality scoring, and candidate deduplication.

Machine Learning Intern (Remote) | Cognifyz Technology

Jan 2025 – Feb 2025 — [Link](#)

- Developed a rating prediction model achieving an R^2 score of 0.98 using preprocessing and feature engineering.
- Built recommendation and classification systems using Jaccard similarity, encoding, normalization, and performance optimization.

Projects

NLP Text Summarization — [Link](#)

- Built an end-to-end text summarization pipeline using the Pegasus Transformer to generate human-like summaries for long documents.
- Automated data ingestion, validation, model training, and experiment tracking using DVC and MLflow.
- Created a backend using FastAPI for the purpose of serving and testing text summarization in real time and allowing inference and validation of the machine learning pipeline through APIs.

Medical AI Chatbot — [Link](#)

- Built an AI medical chatbot using The Gale Encyclopedia of Medicine as a domain-specific knowledge base for accurate context-driven health Q&A.
- Designed and put in place a full Retrieval-Augmented Generation pipeline: extracted medical text, chunked, generated semantic embeddings, stored them in a Pinecone vector database to enable fast similarity search.
- Integrated LangChain with an LLM (FLAN-T5 / Open API) and developed a Flask-based web application to retrieve relevant context and generate grounded responses in real time for user queries.

Drishti — AI-Powered Bilingual Assistant for the Visually Impaired (Capstone)

- Developed a multimodal AI assistive system integrating Whisper, GPT-4o-mini, OpenCV, and real-time audio feedback to help visually impaired users interact through voice commands and scene understanding.
- Built a low-cost prototype (~Rs. 7K) with contextual memory and bilingual interaction capabilities, offering features similar to commercial assistive smart-glasses systems priced between Rs. 1.5L–3.5L.

Technical Skills

Languages: Python, C++, SQL

AI/ML, NLP & GenAI: Pandas, NumPy, Scikit-learn, Regression, Classification, Clustering, Random Forest, KNN, Transformers, Hugging Face, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Whisper, GPT-4o-mini, NLTK, Semantic Search, Vector Embeddings, MLflow, DVC

Backend, APIs, Databases & Deployment: FastAPI, Flask, REST APIs, JWT Authentication, Session Management, API Integration, MySQL, PostgreSQL, Pinecone, Docker, Git, GitHub, OOPs, DBMS, Software Engineering

Certificates

- Deloitte Australia (Forage)** — Data Analytics Job Simulation — [Link](#)
- Tata (Forage)** — GenAI Data Analytics Job Simulation — [Link](#)